DYP DPU

Dr. D. Y. Patil Arts, Commerce & Science College, Pimpri

Academic year 2025-2026

S.Y. B.Sc. (Data Science) Sem IV

Data Structure – II

Question Bank

Chapter 1: Tree	1. Define the term - degree of a tree, left skewed binary tree and right skewed binary tree, expression tree, full binary tree, complete Binary tree, BST. 2. Define max heap, min heap, Terminal node, depth of node, root node, internal node, external node, ancestor 3. Explain the ways of tree representation. 4. Explain tree traversal techniques: i) Inorder ii) preorder iii) Postorder. 5. Traverse the binary tree using given traversal technique i) Inorder ii) preorder iii) Postorder. 6. Construct BST 7. List Applications of tree.
Chapter 2: Efficient Search Trees	1. What is height balance tree? 2. Define Balanace factor 3. List AVL tree Rotations. 4. Explain AVL Tree rotations. 5. Explain concept of Red - Black Tree. 6. Write a note on splay tree. 7. Compare B tree & B+ tree. 8. Write a note on B tree. 9. Construct AVL tree. 10. What is Red Black Tree. State its properties. 11. What is multiday search tree
Chapter 3: Graph	1. List applications of graph. 2. What is weighted graph? 3. Study all graph terminologies 4. BFS and DFS use which data structure for implementation. 5. What is topological sorting? 6. List any two minimum spanning tree algorithms. 7. What is level order traversal? 8. Define indegree & outdegree of vertex with example. 9. Differentiate between DFS & BFS. 10. Write a note on minimum spanning tree. 11. Construct Give i) DFS Traversal ii) BFS Traversal. 12. Compare the data structures: Tree & Graph. 13. Explain Graph representation techniques. 14. Consider following adjacency matrix 1110 0110 1011 0001 i) Draw the graph ii) Give adjacency list 15. Consider given adjacency matrix and i) Draw the graph ii) Give adjacency list 16. Write a note on minimum spanning tree using Kruskal's algorithm and Prims algorithm.
Chapter 4: Hash Table	1. Define Bucket, collision, hashing, hash function, 2. Explain open addressing concept in hash table. 3. Explain various Hash functions. 4. Explain Collision resolution techniques 5. Explain the concept of hashing & rehashing in Hash table. 6. Explain any two properties of good hash function. 7. Write note on linear probing and quadratic probing. 8. Explain chaining in hashtable.

Also study all practical assignmnets